Topic 7: Acids, bases and salts - Answers

IGCSE Chemistry 0620 Paper 4 topical answer booklet.

Subtopic: Acids, bases, salts and pH

Questions in this answer PDF: 23

Use this with the matching topical question PDFs. The answer points are organised by the same source labels.

Short explanation notes are added for harder calculation, electrolysis, equilibrium, rate, salts, organic, metals and practical questions.

Periodic table reference pages are not included to save printing paper.

2020 May/June Paper 41 Question 4

Main topic: Topic 7: Acids, bases and salts | Marks: 16 | Answer source: 0620_s20_ms_41.pdf

Extra topic(s): Topic 3: Stoichiometry, Topic 12: Experimental techniques and chemical analysis

Answer points

4(a)

- (damp) litmus
- (turns) blue

4(b)(i)

- proton acceptor

4(b)(ii)

- Above pH 7 up to 11

4(b)(iii)

- blue precipitate
- precipitate dissolves
- deep blue solution remains

4(c)(i)

- neutralisation

4(c)(ii)

- Na_2SO_4
- $2H_2O$

4(d)(i)

- methyl orange

4(d)(ii)

- mol of NaOH
- (
-)
- 25.0
- 0.0400
- 0.001 00
- =
- x
- =
- mol
- mol of H_2SO_4
- (
-)
-
- 0.001
- 0.0005 00
- =
- =
- =
- 1000
- 0.0005
- 0.025
- 20.0
- 20.0
- x

- =
- x
- =
- (mol / dm3)
- error carried forward may be allowed

4(d)(iii)

- use of 40 g/mol
- 40 x 0.04 = 1.6 (g/dm3)

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2020 Oct/Nov Paper 41 Question 3

Main topic: Topic 7: Acids, bases and salts | Marks: 13 | Answer source: 0620_w20_ms_41.pdf

Answer points

3(a)(i)

- 2 -> 2 + 2

3(b)

- 75(%)

3(c)

- test: (damp red) litmus paper (1)
- result: (litmus goes) blue (1)

3(d)(i)

- diffusion

3(d)(ii)

- particles move from an area of high to low concentration
- particles move randomly

3(d)(iii)

- CO2 molecules are heavier (than NH3)

3(e)(i)

- lower temperature: (rate of reaction) slower (1)
- higher pressure: expensive/specialist equipment

3(e)(ii)

- catalyst

3(f)(i)

- proton acceptor

3(f)(ii)

- any value greater than 7 up to 12

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not escape.*

2020 Oct/Nov Paper 42 Question 2

Main topic: Topic 7: Acids, bases and salts | Marks: 26 | Answer source: 0620_w20_ms_42.pdf

Answer points

2(a)

- HNO3

2(b)(i)

- to make sure all the (sulfuric) acid reacts

2(b)(ii)

- no (more) fizzing (1)
- (FeCO3) stops dissolving or a solid remains / is visible (in the mixture) (1)

2(b)(iii)

- rinse the residue (with distilled water)

2(b)(iv

- a solution that can dissolve no more solute (1)
- at the specified temperature (1)

2(b)(v)

- iron(II) oxide / iron(II) hydroxide

2(c)

- mass of $FeSO_4$ = 152 (1)
- mass of H_2O = 278 - 152 = 126 (1)
- mol of H_2O = 126 / 18 and x = 7 (1)

2(d)(i)

- precipitation

2(d)(ii)

- cream precipitate

2(d)(iii)

- $Ag^+(aq) + Br^-(aq) \rightarrow AgBr(s)$
- AgBr (as only product) (1)
- Ag^+ and Br^- (as reactants)(1)
- state symbols(1)

2(e)

- mol of NaCl = 2.34 / 58.5 = 0.04(00)
- mol of Cl 2 = /2 = 0.04(00)/2 = 0.02(00)
- 0.02(00) x 24000 = 480 (cm^3)

2(f)(i)

- ions (1)
- (ions) are fixed (in a lattice) (1)
- ions are mobile (1)

2(f)(ii)

- chlorine

2(f)(iii)

- oxidation (1)
- electrons are lost (1)

2(f)(iv)

- dissolve it (in water)

Easy explanation / reminder

- *Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.*
- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2021 Feb/March Paper 42 Question 4

Main topic: Topic 7: Acids, bases and salts | Marks: 17 | Answer source: 0620_m21_ms_42.pdf
Extra topic(s): Topic 3: Stoichiometry, Topic 12: Experimental techniques and chemical analysis

Answer points

4(a)

- $Zn(s) + 2HCl(aq) \rightarrow ZnCl_2(aq) + H_2(g)$
- $ZnCl_2$ or H_2 as a product (1)
- correct equation (1)
- states (1)

4(b)

- to make sure all the (hydrochloric) acid reacts

4(c)

- filtration

4(d)

- a solution that can dissolve no more solute (1)
- at a given temperature (1)

4(e)

- solubility (of $ZnCl_2$ / solids) decreases (as temperature decreases)

4(f)

- zinc oxide
- zinc carbonate

4(g)

- Ca will also react with water

4(h)(i)

- neutralisation

4(h)(ii)

- 0.100 x 25 / 1000 = 0.0025(0) (1)
- 0.0025 x 2 = 0.005(00) (1)
- 0.005 x 1000 / 20 = 0.25(0) (1)
- Mr = 40 (1)
- 0.25 x 40 = 10.(0) (1)

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2021 May/June Paper 41 Question 1

Main topic: Topic 7: Acids, bases and salts | Marks: 6 | Answer source: 0620_s21_ms_41.pdf

Answer points

1(a)

- Haber (process)

1(b)

- fractional distillation

1(c)

- electrolysis

1(d)

- filtration

1(e)

- hydrolysis

1(f)

- chromatography

Easy explanation / reminder

- *Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.*
- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2021 May/June Paper 41 Question 5

Main topic: Topic 7: Acids, bases and salts | Marks: 15 | Answer source: 0620_s21_ms_41.pdf

Extra topic(s): Topic 3: Stoichiometry

Answer points

5(a)

- (add) water (to both salts) (1)
- dissolve both salts / make solutions (1)
- filter (lead(II) iodide)(1)
- wash (residue of lead(II) iodide) with water AND dry e.g. with filter paper / description of washing and drying (1)
- Pb(NO3)2 + 2 NaI -> 2NaNO3 + PbI2
- OR Pb2+ + 2I- -> PbI2 (1)

5(b)(i)

- glowing splint (1)
- relights / rekindles (1)

5(b)(ii)

- 2ZnO(s) and 4NO2(g) (1)
- 12H2O(g) (1)

5(c)(i)

- heat again and weigh again / repeat steps 2 and 3 (1)
- until mass is constant (1)

5(c)(ii)

- 0.005 (1)
- 0.9 (1)
- (0.9 ÷ 18 =) 0.05 (1)
- (0.05 ÷ 0.005 =) 10 (1)

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2021 May/June Paper 43 Question 5

Main topic: Topic 7: Acids, bases and salts | Marks: 15 | Answer source: 0620_s21_ms_43.pdf
Extra topic(s): Topic 3: Stoichiometry

Answer points

5(a)

- add zinc carbonate to sulfuric acid until
- -
- it stops dissolving
- OR
- -
- no more effervescence (1)
- filter (zinc carbonate) (1)
- evaporation of filtrate to form dry crystals (1)
- ZnCO3 + H2SO4 -> ZnSO4 + CO2 + H2O
- CO2 product in equation (1)
- rest of equation correct (1)

5(b)(i)

- acidified aqueous potassium manganate(VII) (1)
- purple to colourless (1)

5(b)(ii)

- 2 FeSO4.7H2O (1)
- 14 H2O (1)

5(c)(i)

- heat + weigh + repeat (1)
- until mass is constant (1)

5(c)(ii)

- 0.02 (1)
- 0.72 (1)
- ÷ 18 = 0.72/18 = 0.04 (1)
- ÷ = 0.04 ÷ 0.02 = 2 (1)
- Question Answer

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2021 Oct/Nov Paper 42 Question 2

Main topic: Topic 7: Acids, bases and salts | Marks: 10 | Answer source: 0620_w21_ms_42.pdf
Extra topic(s): Topic 3: Stoichiometry

Answer points

2(a)(i)

- strong

2(a)(ii)

- 2H+ + SO42-
- H+ (1) correct equation (1)

2(a)(iii)

- pink / red

2(b)(i)

- ZnCO3(s) + 2HNO3(aq) ->
- Zn(NO3)2(aq) + H2O(l) + CO2(g)
- reactant states (1) product states (1)

2(b)(ii)

- 125
- 2.5 / 125 = 0.02(00)

- 0.02(00) x 2 = 0.04(00)
- 0.04(00) x 1000 / 20 = 2(.00)

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2021 Oct/Nov Paper 43 Question 5

Main topic: Topic 7: Acids, bases and salts | Marks: 11 | Answer source: 0620_w21_ms_43.pdf
Extra topic(s): Topic 12: Experimental techniques and chemical analysis

Answer points

5(a)(i)

- too many colour changes

5(a)(ii)

- Any acid-base indicator, e.g. methyl orange or phenolphthalein

5(b)

- repeat without indicator using same volumes OR remove indicator by adding charcoal or carbon and filtering (1)
- evaporate / heat / warm/
- boil/leave in hot place (1)
- until most of the water is gone / some water left /
- saturation(point) /
- crystallisation (point) /
- evaporate some of the water (1)
- cool /
- leave to crystallise(1)
- description of drying (1)

5(c)(i)

- yellow flame

5(c)(ii)

- solid dissolves / disappears(1)
- blue solution(1)

5(c)(iii)

- white precipitate

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2022 May/June Paper 41 Question 6

Main topic: Topic 7: Acids, bases and salts | Marks: 11 | Answer source: 0620_s22_ms_41.pdf
Extra topic(s): Topic 9: Metals

Answer points

6(a)(i)

- zinc blende

6(a)(ii)

- heat(zinc sulfide) strongly in air / roast in air

6(a)(iii)

- carbon or carbon monoxide

6(a)(iv)

- the temperature in the furnace is above or higher than the boiling point of zinc
- OR
- the boiling point of zinc is below or less than the temperature of the furnace

6(a)(v)

- condensation / condensing / condense

6(b)(i)

- no bubbles
- or
- no fizzing

- or
- no effervescence

6(b)(ii)

- zinc / zinc oxide / zinc hydroxide

6(b)(iii)

- (powder has) larger surface area OR lumps have smaller surface area (1)
- (powder has) more collisions per unit time / more collision frequency
- OR
- lumps have fewer collisions per unit time / less collision frequency (1)

6(b)(iv)

- crystals (form on glass rod or microscope slide)

6(b)(v)

- ZnSO4

Easy explanation / reminder

- *Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than activation energy.*
- *Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.*

2022 Oct/Nov Paper 41 Question 2

Main topic: Topic 7: Acids, bases and salts | Marks: 23 | Answer source: 0620_w22_ms_41.pdf

Extra topic(s): Topic 3: Stoichiometry, Topic 2: Atoms, elements and compounds

Answer points

2(a)

- metallic (1)
- lattice of potassium ions (1)
- sea of electrons (1)
- attraction between potassium ions and electrons (1)

2(b)(i)

- any two (one from each bullet point)
- -
- physical constants: high boiling point / melting point
- -
- conductivity: conduct electricity when aqueous / conduct electricity when molten
- -
- solubility: soluble in water

2(b)(ii)

- eight dots in third shell of both K (1)
- six crosses and two dots in third shell of S (1)
- ‘+' charge on each K on correct answer line and ‘2-‘ charge on S ion on correct answer line (1)

2(c)(i)

- lilac

2(c)(ii)

- OH-

2(c)(iii)

- blue

2(c)(iv)

- mol of K = 2.34 / 39 = 0.06(00) (1)
- mol of H2 = 0.06 / 2 = 0.03(00) (1)
- volume of H2 = 0.03 -> 24 000 = 720 cm3 (1)

2(d)(i)

- hydrochloric (acid)

2(d)(ii)

- neutralisation

2(d)(iii)

- titration

2(e)(i)

- white

2(e)(ii)

- silver chloride

2(e)(iii)
- $Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$
- AgCl (as only product) (1)
- Ag^+ and Cl^- (as only reactants) (1)
- state symbols (1)

Easy explanation / reminder
- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.*

2023 May/June Paper 41 Question 6

Main topic: Topic 7: Acids, bases and salts | Marks: 13 | Answer source: 0620_s23_ms_41.pdf
Extra topic(s): Topic 10: Chemistry of the environment

Answer points
6(a)(i)
- $4FeS_2 + 11O_2 \rightarrow 2Fe_2O_3 + 8SO_2$

6(a)(ii)
- iron(III) oxide

6(b)(i)
- yield of SO_3 is less

6(b)(ii)
- yield of SO_3 is less
- OR
- rate is less

6(c)
- $2NH_3 + H_2SO_4 \rightarrow (NH_4)_2SO_4$
- $(NH_4)_2SO_4$ on the right (1)
- equation correct(1)

6(d)(i)
- lead(II) nitrate

6(d)(ii)
- $Pb^{2+}(aq) + SO_4^{2-}(aq) \rightarrow PbSO_4(s)$
- $PbSO_4$ on the right(1)
- only Pb^{2+} and SO_4^{2-} on the left(1)
- (aq) + (aq) -> (s)(1)

6(d)(iii)
- filter(1)
- wash (the residue or lead sulfate) with distilled or deionised water(1)
- description of drying(1)

Easy explanation / reminder
- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not escape.*

2023 May/June Paper 42 Question 4

Main topic: Topic 7: Acids, bases and salts | Marks: 21 | Answer source: 0620_s23_ms_42.pdf
Extra topic(s): Topic 3: Stoichiometry

Answer points
4(a)
- proton acceptor

4(b)
- alkali

4(c)
- blue

4(d)
- sodium chloride
- water

- ammonia

4(e)(i)

- amphoteric (oxides)

4(e)(ii)

- aluminium oxide
- or
- zinc oxide

4(f)(i)

- all single bonding dot and cross pairs correct
- double C = O bond dot and cross pairs are correct
- complete diagram is correct

4(f)(ii)

- 3<= pH < 7

4(f)(iii)i

- CH3COOH ■ CH3COO- + H+
- H+
- CH3COO-
- use of ■

4(f)(iv)

- H+ + OH- ■ H2O

4(g)

- mol KOH = 0.0800 -> 25 / 1000
- = 0.002(00) / 2(00) -> 10-3
- mol H2SO4 = / 2 = 0.002 / 2
- = 0.001(00) / 1(00 -> 10-3
- = -> 1000 / 20 = 0.001 -> 1000 / 20
- = 0.05(00) / 5.(00) -> 10-2
- = 98
- = 98 -> = 98 -> 0.05(00) = 4.9(0) (g / dm3)

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2023 Oct/Nov Paper 42 Question 4

Main topic: Topic 7: Acids, bases and salts | Marks: 14 | Answer source: 0620_w23_ms_42.pdf

Extra topic(s): Topic 4: Electrochemistry, Topic 12: Experimental techniques and chemical analysis, Topic 2: Atoms, elements and compounds

Answer points

4(a)(i)

- lead nitrate (1)
- soluble chloride e.g. sodium chloride (1)

4(a)(ii)

- Pb2+(aq) + 2Cl-(aq) -> PbCl2(s)
- PbCl2 as only product (1)
- Pb2+ + 2Cl- as only reactants (1)
- state symbols (1)

4(a)(iii)

- filter (1)
- wash (the residue) using water (1)
- dry the residue between filter papers / in a warm place (1)

4(b)(i)

- mobile ions

4(b)(ii)

- 2Cl- -> Cl2 + 2e-
- any negative Cl species losing electron(s) (1)
- correct ionic half equation (1)

4(b)(iii)

- (damp) litmus (paper) (1)
- is bleached / goes white (1)

- 4biv
- (shiny) grey AND solid

Easy explanation / reminder

- Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

2023 Oct/Nov Paper 43 Question 1

Main topic: Topic 7: Acids, bases and salts | Marks: 6 | Answer source: 0620_w23_ms_43.pdf

Extra topic(s): Topic 11: Organic chemistry, Topic 9: Metals, Topic 12: Experimental techniques and chemical analysis, Topic 10: Chemistry of the environment

Answer points

1(a)

- potassium iodide

1(b)

- sodium bromide

1(c)

- sulfur dioxide

1(d)

- propene

1(e)

- carbon monoxide

1(f)

- anhydrous copper(II) sulfate

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

- Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO2 and H2O when oxygen is in excess.

2023 Oct/Nov Paper 43 Question 4

Main topic: Topic 7: Acids, bases and salts | Marks: 15 | Answer source: 0620_w23_ms_43.pdf

Extra topic(s): Topic 3: Stoichiometry

Answer points

4(a)(i)

- dilute sulfuric acid

4(a)(ii)

- solid remains / solid undissolved (1)
- bubbling / effervescence / fizzing stops (1)

4(a)(iii)

- aqueous zinc sulfate

4(a)(iv)

- any two
- -
- zinc oxide
- -
- zinc carbonate
- -
- zinc hydroxide

4(b)(i)

- (a solution containing the) maximum amount of solute dissolved / no more solute can dissolve (1)
- at a given temperature (1)

4(b)(ii)

- solubility decreases as temperature decreases

4(c)(i)

- hydrated

4(c)(ii)

- to ensure all the water of crystallisation is removed

4(c)(iii)

- 2 -> 10-3 / 0.002 (1)

- 0.144 (g) (1)
- (0.144 ■ 18 =) 8 -> 10-3 / 0.008 (1)
- (0.008 ■ 0.002 =) 4 (1)

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2024 Feb/March Paper 42 Question 3

Main topic: Topic 7: Acids, bases and salts | Marks: 16 | Answer source: 0620_m24_ms_42.pdf

Extra topic(s): Topic 3: Stoichiometry

Answer points

3(a)

- proton acceptor

3(b)

- a soluble base

3(c)

- blue
- colourless

3(d)(i)

- HNO3
- lowest pH

3(d)(ii)

- universal indicator

3(e)

- (CH3COOH) ■ CH3COO- + H+
- H+
- CH3COO-
- ■
- February/March 2024

3(f)

- H+ + OH- -> H2O

3(g)

- (0.0150 -> 20.0/1000 =) 0.0003(00) / 3.00 -> 10-4 (mol)
- (-> 2 = 3.00 -> 10-4 -> 2 =) 0.0006(00) / 6.00 -> 10-4 (mol)
- (-> 1000/25.0 = 6.00 -> 10-4 -> 1000 / 25.0 =) 0.0240 (mol / dm3)
- 63 (g / mol)
- (-> = 0.0240 -> 63 =) 1.51(2) (g / dm3)

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2024 May/June Paper 43 Question 5

Main topic: Topic 7: Acids, bases and salts | Marks: 17 | Answer source: 0620_s24_ms_43.pdf

Extra topic(s): Topic 3: Stoichiometry

Answer points

5(a)(i)

- named soluble barium salt e.g. barium chloride / barium nitrate
- named soluble sulfate salt
- e.g. sodium sulfate / potassium sulfate / ammonium sulfate / sulfuric acid

5(a)(ii)

- filter
- wash (residue) with water
- and
- dry (residue) between filter papers / in a warm place

5(a)(iii)

- Ba2+ (aq) + SO42- (aq) -> BaSO4(s)

- $BaSO_4$ as only product
- Ba^{2+} + SO_4^{2-} as only reactants
- state symbols

5(b)
- sulfuric (acid)
- copper(II) carbonate / copper(II) oxide / copper(II) hydroxide

5(c)(i)
- glowing splint and relights

5(c)(ii)
- $2CuO + 4NO_2$
- $6H_2O$

5(d)(i)
- (To ensure that) all the water is given off

5(d)(ii)
- mol $ZnSO_4$
- (0.322 / 161 =) 0.002(00)
- mass H_2O given off
- (0.574 - 0.322 =) 0.252
- mol H_2O given off
- (0.252 ÷ 18 =) 0.014(0)
- value of x
- (0.014 ÷ 0.002 =) 7

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2024 Oct/Nov Paper 41 Question 1

Main topic: Topic 7: Acids, bases and salts | Marks: 5 | Answer source: 0620_w24_ms_41.pdf
Extra topic(s): Topic 6: Chemical reactions

Answer points

1(a)
- D

1(b)
- F

1(c)
- C

1(d)
- B AND C

1(e)
- B

Easy explanation / reminder

- *Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not escape.*
- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2024 Oct/Nov Paper 41 Question 3

Main topic: Topic 7: Acids, bases and salts | Marks: 18 | Answer source: 0620_w24_ms_41.pdf
Extra topic(s): Topic 4: Electrochemistry

Answer points

3(a)
- H_2O
- state symbols: (s) (aq) (aq)

3(b)
- no more solid dissolves

3(c)
- filtration

3(d)(i)

- substance that is chemically combined with water

3(d)(ii)

- $CuSO_4 \cdot 5H_2O$

3(d)(iii)

- heat
- until saturation (point) AND allow to cool

3(e)(i)

- mobile ions

3(e)(ii)

- conducts electricity
- inert

3(e)(iii)

- becomes lighter (blue)

3(e)(iv)

- pink AND solid

3(e)(v)

- $4OH^- \rightarrow 2H_2O + O_2 + 4e^-$
- O_2 as a product
- OH^- AND e^-
- correct equation

3(e)(vi)

- one mark each for any two of:
- -
- colour remains constant
- -
- no bubbles at the anode
- -
- anode dissolves

Easy explanation / reminder

- Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

2024 Oct/Nov Paper 43 Question 4

Main topic: Topic 7: Acids, bases and salts | Marks: 20 | Answer source: 0620_w24_ms_43.pdf

Extra topic(s): Topic 3: Stoichiometry, Topic 12: Experimental techniques and chemical analysis

Answer points

4(a)(i)

- blue (1)
- to
- colourless (1)

4(a)(ii)

- too many colour changes

4(b)

- 0.0025 / 2.5 -> 10-3 (mol)
- 0.00125 / 1.25 -> 10-3 (mol)
- 0.0625 / 6.25 -> 10-2 (mol / dm3)

4(c)

- heat the solution / warm the solution /boil the solution / leave solution in hot place
- to saturation (point)/ crystallisation point AND leave to cool
- suitable method of drying

4(d)

- $KOH + H_2SO_4 \rightarrow KHSO_4 + H_2O$
- 4e(i)
- acid / acidic

4(e)(ii)

- flame test is carried out on X:
- lilac flame
- solid copper(II) carbonate is added to X:
- solid dissolves / solid disappears

- bubbles / fizzing / effervescence
- blue solution
- aqueous barium nitrate acidified with dilute nitric acid is added to X:
- white precipitate

4(f)(i)

- moles of Zn = 0.005 / 5 -> 10-3 (mol)
- Zn is limiting because moles of H2SO4 ■ moles of Zn AND 1:1 ratio required for reaction

4(f)(ii)

- (48.0 ÷ 24 000 =) 0.00200 (mol)
- 1.20 -> 1021 (molecules)

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2025 Oct/Nov Paper 42 Question 4

Main topic: Topic 7: Acids, bases and salts | Marks: 15 | Answer source: 0620_w25_ms_42.pdf

Answer points

4(a)(i)

- H+

4(a)(ii)

- complete (dissociation)
- partial (dissociation)

4(a)(iii)

- ethanoic (acid)

4(a)(iv)

- colourless

4(a)(v)

- SO42-
- CH3COO -

4(a)(vi)

- hydrogen
- oxygen

4(a)(vii)

- calcium ethanoate

4(b)(i)

- proton acceptor

4(b)(ii)

- insoluble

4(b)(iii)

- aluminium oxide

4(b)(iv)

- use of -18 or (9 -> -2) or (3 -> 3 -> -2)
- +5

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.*

2025 Oct/Nov Paper 43 Question 5

Main topic: Topic 7: Acids, bases and salts | Marks: 12 | Answer source: 0620_w25_ms_43.pdf

Extra topic(s): Topic 12: Experimental techniques and chemical analysis, Topic 9: Metals, Topic 8: The Periodic Table

Answer points

5(a)(i)

- Pb2+(aq) + SO42- (aq) -> PbSO4(s)
- PbSO4 as the ONLY formula on the right
- ONLY Pb2+ and SO42- on the left
- state symbols

5(a)(ii)

- lead(II) nitrate

5(a)(iii)

- sodium sulfate / potassium sulfate / ammonium sulfate

5(a)(iv)

- residue

5(a)(v)

- wash with distilled water
- suitable method of drying

5(b)(i)

- any two from:
- -
- good conductor of electricity
- -
- soluble salts
- -
- basic oxides

5(b)(ii)

- any two from:
- -
- variable oxidation states
- -
- coloured compounds
- -
- catalyst

Easy explanation / reminder

- Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than activation energy.

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.